Eric Chen

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Education

University of Southern California — Aug. 2023 — May 2027
M.S. Electrical Engineering | B.S. Computer Engineering & Computer Science — *Los Angeles, CA*

Experience

Technology Intern — May 2025 — Aug. 2025
MUJI USA — *New York, NY*

- Built REST APIs to sync e-commerce orders with the WMS; cut p95 order-sync latency by 42% across NYC stores.
- Optimized inventory/allocations tables and queries, making reports 3.1× faster and cutting p99 latency 61%.
- Built idempotent verification checks; improved online-warehouse consistency and reduced leftover stock issues.
- Wrote daily data integrity scripts (diff scans, duplicates); prevented 1.3k+ bad records/month from reaching reports.

AI Engineer Fellow — Dec. 2025 — Present
Handshake AI — *Remote*

- Fine-tuning large language models for code generation using supervised learning on open-source codebases.
- Curating gold-standard demonstrations emphasizing functional correctness and multi-file reasoning capabilities.
- Evaluating model outputs against benchmarks for accuracy, code quality, and cross-file dependency handling.
- Built CI/CD pipelines and Dockerized environments for autonomous regression testing across model versions.

Course Producer for CS353 (Intro to Networking) — Spring 2026
University of Southern California — *Los Angeles, CA*

- Mentored students in CS353 through weekly tutoring hours with structured feedback on algorithms and code quality.
- Collaborated with course staff to clarify material, debug student code, and improve overall learning outcomes.

Projects

Parallel MCCFR for No-Limit Texas Hold'em — Apr. 2026
CUDA C++, cuRAND, OpenMP, MPI, A100 GPU

- Built a poker AI to approximate Nash equilibrium via Monte Carlo CFR self-play and Linear CFR+ (used by Pluribus).
- Ported to CUDA C++, running 65k simultaneous games per A100 kernel for a 56,000× speedup over the Python baseline.
- Eliminated Python serialization overhead (57s/worker) via shared-memory merging and GPU-optimized data layouts.
- Deployed a live-play bot using hash-based strategy lookup for sub-microsecond decision latency.

Game Resource Detection System — Jul. 2025
Python, PyTorch, YOLOv11, OpenCV

- Built and annotated a custom dataset in Roboflow to train YOLOv11 for detecting in-game resources.
- Applied preprocessing and augmentation with OpenCV and NumPy to improve detection robustness.
- Trained YOLOv11 in PyTorch, achieving real-time performance; deployed as in-game overlay for automation.

Makeshift Router — Apr. 2025
C++, POSIX Threads, Sockets, Network Protocol Design

- Constructed a multi-threaded router simulation handling concurrent packet forwarding, routing, and TTL expiration.
- Designed a layered protocol stack with checksums, distance-vector routing, and dynamic table updates.
- Implemented socket-based IPC for inter-node communication with custom serialization and timeout handling.

Rankify — Dec. 2024
React, Node.js, Java, SQL, WebSockets

- Developed a social web app integrating the Spotify API for users to rank songs and interact with friends.
- Implemented song previews, rating system, user profiles, and leaderboards; added real-time chatrooms.

Technical Skills

Languages: Python, C++, Java, JavaScript/TypeScript, SQL, Verilog
Frameworks/Libraries: React, Node.js/Express, Flask, PyTorch, CUDA, OpenCV, NumPy, WebSockets
Systems/Tools: Linux, Git, Docker, GDB, OpenMP, POSIX Threads, Sockets, Vivado, VS Code, Cadence, Wireshark
Databases/Cloud: PostgreSQL, MySQL, Google Cloud Platform
Other: AutoCAD (Certified) — Mandarin (fluent)

Relevant Coursework

Data Structures and Object-Oriented Design; Principles of Software Development; Introduction to Algorithms and Theory of Computing; Introduction to Operating Systems; Computer System Organization; Introduction to Digital Circuits; Professional C++; Parallel and Distributed Computing; MOS VLSI Circuit Design